

```
In [2]: from geof_interp_widget import (
        ESCALAS, build_selector_widget, interpolate_mosaic_and_update,
        export_geotiff_stack, save_all_stacks, run_workflow_quick
    )

    # (A) Interface com widgets:
    widgets = build_selector_widget() # carrega dados via botão; variable global 'quadricula' é atualizada
```

```
/home/ggrrl/.pyenv/versions/3.13.7/envs/geologia/lib/python3.13/site-packages/pyogrio/raw.py:198: RuntimeWarning: driver GPKG does not support open option DRIVER
    return ogr_read(
VBox(children=(HBox(children=(Dropdown(description='Escala', index=2, options=('25k', '50k', '100k', '250k', '50k', '...
```

```
In [5]: # (B) Fluxo rápido sem widgets:
mc, quadricula = run_workflow_quick(
    escala="25k",
    ids=["SF23_YB_I1_NW"],
    gama="gama_line_1105", mag="mag_line_1105", extend_size=600
)

# Interpola no mosaico (com buffer) e recorta por folha (sem costura):
value_fields = { "gama_line_1105": "TC", "mag_line_1105": "TMI" }
quadricula = interpolate_mosaic_and_update(
    quadricula, value_fields, folhas_gdf=mc, epsg=mc.crs.to_epsg() if mc.crs else None,
    spacing=100.0, reduce_spacing=500.0, buffer_m=600.0, dtype="float32", nodata=float("nan")
)

# Exporta 1 GeoTIFF por folha (bandas nas ordem de 'canais'):
save_all_stacks(quadricula, canais=["gama_line_1105", "mag_line_1105"], outdir="out_tif")
```

```
/home/ggrrl/.pyenv/versions/3.13.7/envs/geologia/lib/python3.13/site-packages/pyogrio/raw.py:198: RuntimeWarning: driver GPKG does not support open option DRIVER
    return ogr_read(
# Montando grade...
```

```
lit [00:00, 4843.31it/s]
- Folha "SF23_YB_I1_NW" adicionada.
```

```
1 folhas adicionadas.
# Carregando geofísica...
```

```
100%|████████████████████████████████████████████████████████████████████████████████| 1/1 [00:00<00:00, 12.68it/s]
- gama_line_1105 atualizado na folha: SF23_YB_I1_NW com 5168 pontos
- mag_line_1105 atualizado na folha: SF23_YB_I1_NW com 56273 pontos
```

```
100%|████████████████████████████████████████████████████████████████████████████████| 1/1 [00:00<00:00, 20560.31it/s]
```

```
/home/ggrrl/.pyenv/versions/3.13.7/envs/geologia/lib/python3.13/site-packages/pyogrio/raw.py:198: RuntimeWarning: driver GPKG does not support open option DRIVER
    return ogr_read(
Folhas ativas: 1
```

```
[interp] Canal: gama_line_1105 • valor='TC'
[info] Canal 'gama_line_1105': coluna preferida='TC', usadas=['MDT']
```

```

-----
ValueError                                Traceback (most recent call last)
Cell In[5], line 10
      8 # Interpola no mosaico (com buffer) e recorta por folha (sem costura):
      9 value_fields = { "gama_line_1105": "TC", "mag_line_1105": "TMI" }
--> 10 quadricula = interpolate_mosaic_and_update(
11     quadricula, value_fields, folhas_gdf.mc, epsg=mc.crs.to_epsg() if mc.crs else None,
12     spacing=100.0, reduce_spacing=500.0, buffer_m=600.0, dtype="float32", nodata=float("nan")
13 )
14
15 # Exporta 1 GeoTIFF por folha (bandas nas ordem de 'canais'):
16 save_all_stacks(quadricula, canais=["gama_line_1105", "mag_line_1105"], outdir="out_tif")

File ~/projetos/PreditorTerra/geof_interp_widget.py:461, in interpolate_mosaic_and_update(quadricula, c
anais, folhas_gdf, epsg, spacing, reduce_spacing, damping, mindist, buffer_m, dtype, nodata)
458 X, Y, Z = _collect_xyz(quadricula, canal, value_col)
460 reducer = vd.BlockReduce(reduction=np.median, spacing=reduce_spacing)
--> 461 xr, yr, zr = reducer.filter(X, Y, Z)
462 print(f" pontos: {len(Z):,} → reduzidos: {len(zr):,}")
464 spline = vd.Spline(damping=damping, mindist=mindist).fit((xr, yr), zr)

File ~/.pyenv/versions/3.13.7/envs/geologia/lib/python3.13/site-packages/verde/blockreduce.py:160, in B
lockReduce.filter(self, coordinates, data, weights)
117 def filter(self, coordinates, data, weights=None): # noqa: A003
118     """
119     Apply the blocked aggregation to the given data.
120
121     (...)
122     158
123     159
124     """
--> 160 coordinates, data, weights = check_fit_input(
161     coordinates, data, weights, unpack=False
162 )
163 blocks, labels = block_split(
164     coordinates,
165     spacing=self.spacing,
166     region=self.region,
167 )
168
169 if any(w is None for w in weights):
170     raise ValueError("Data arrays must have the same shape () as coordinates. Data shapes: [(5168,)].")

File ~/.pyenv/versions/3.13.7/envs/geologia/lib/python3.13/site-packages/verde/base/coords.py:240, in ch
eck_fit_input(coordinates, data, weights, unpack)
238 coordinates = check_coordinates(coordinates)
239 if any(i.shape != coordinates[0].shape for i in data):
--> 240     raise ValueError(
241         "Data arrays must have the same shape {} as coordinates. Data shapes: {}".format(
242             coordinates[0].shape, [i.shape for i in data]
243         )
244     )
245 if any(w is not None for w in weights):
246     if len(weights) != len(data):
247         raise ValueError("Data arrays must have the same shape () as coordinates. Data shapes: [(5168,)].")

```

In []: